A college conducted a survey to explore academic interests of students. If asked students to place checks besides the following statements if they were true:

1. I was on the honor roll last term.
2. I belong to an academic club.
3. I'm majoring in multiple subjects.

Out of 100 students, 28 checked #1, 26 checked #2, and 14 checked #3, #8 checked both #1 and #2, 4 checked #1 and #3, 3 checked #2 and #3, and 2 checked all three.

- How many checked at least one? $N(H \wedge C \wedge \wedge D) = N(H) + N(C) + N(D) - N(N \wedge C) - N(H \wedge D) - N(C \wedge D) + N(H \wedge C \wedge D) =$ **28 + 26 + 14 - 8 - 4 - 3 + 2** $=$ **55**
- How many checked none? **100 - 55 = 45**
- Let $H$ be the set of students who checked #1, $C$ those who checked #2, and $D$ those who checked #3. *Insert Picture*
- How many checked #1 and #2, but not #3. $N(H \wedge C) - N(H \wedge C \wedge D) =$ **8 - 2 = 6**
- How many checked #2 and #2, but not #1. $N(C \wedge D) - N(C \wedge D \wedge H) =$ **3 - 2 = 1**
- How many checked #2, but neither of the others? $N(C) - N(C \wedge H) - N(C \wedge D) + N(C \wedge D) + N(C \wedge H \wedge D) =$ **26 - 8 - 3 + 2 = 17**

# 1 8.4 The Pigeonhole Problem

- If 13 cards are selected from a standard 52-card deck, must at least 2 be of the same denomination? No, example: $2, 3, 4, \ldots, 10, J, Q, K, A$.
- If 20 cards are selected from a standard 52-card deck, must at least 2 be of the same denomination? Yes, let $X$ be the set consisting of the 20 selected cards and let $Y$ be the 13 possible denominations. Define a function D from X (pigeons) to Y (Pigeonholes) by specifying $\forall x \in X, D(X) = The denomination of X$. Now $X$ has 20 elements and $Y$ has 13, and $20 > 13$. So by the pigeonhole principle, D is not one-to-one. Hence, $\exists x_1, x_2, x_1 \neq x_2 \wedge D(x_1) = D(x_2)$. Then $x_1$ and $x_2$ are distinct cards that have the same denomination.

### Example 4.27

In a group of 2000 people, must at least 5 have the same BD? Yes, let $X$ be the set consisting of the 2000 people (Pigeons) and $Y$ be the set of 366 days (Pigeon Holes). Define a function $B : X \to Y$ by specifying that $\forall x \in X, B(x) = x$'s birthday. Now $2000 > 4 \times 366 = 1464$ and so by the generalized pigeonhole principle, there must be some birthday $y \in Y$ such that $B^{-1}(y$ has at least $4 + 1 = 5$ elements. Hence at least 5 people must share the same birthday.

### Example 4.30

- $12 \times 1967$ Penny
- $7 \times 1968$ Penny
- $11 \times 1971$ Penny

How many must you pick to have at least 5 from the same year? **13**.

## 9.5 Counting Subsets of a Set: Combinations

### Theorem 9.5.1

The number of subsets of size r, called r - combinations, that can be chosen from a set of n elements, $\binom{n}{r}$ is.

$$\binom{n}{r} = \frac{P(n,r)}{r!} = \frac{n!}{r!(n-r)!} \tag{1}$$

### Example 9.5.9

$\binom{40}{6} = \frac{40!}{6!34!} = 3,838,380$

### Example 9.5.21

How many Morse symbols with 7 or fewer dots or dashes? $2^1 + 2^2 + 2^3 + 2^4 + \cdots + 2^7 = 2 \times \frac{2^7-1}{2-1} = 254$